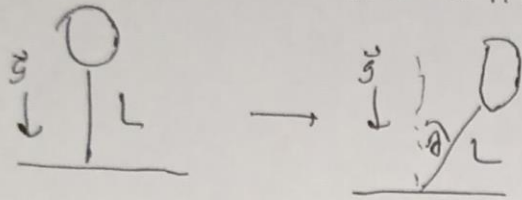


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Jawaban Modul 7 (no 3, 7, 12, 14, dan 15)

3.



$$P_{he} = 0,179 \text{ kg} \cdot \text{m}^3$$

$$g = 10 \text{ m/s}^2$$

$$P_{ud} = 1,20 \text{ kg} \cdot \text{m}^3$$

$$L = 3,00 \text{ m}$$

(a) Menunjukkan GHS $\Rightarrow$ gaya pemulih / torsi sebanding dengan negatif simpangan, yaitu $T \propto -\theta \Rightarrow \alpha \propto -\theta$ (dari $\tau = I\alpha$)

$F_{net} = F_{apung} - W$ karena $P_{ud} > P_{he}$ sehingga F_{net} ke atas

$$= P_{ud} \cdot V \cdot g - P_{he} \cdot V \cdot g$$

$$= (P_{ud} - P_{he}) \cdot V \cdot g$$

Tinjau torsi, yaitu $\tau = F \cdot r \Rightarrow \tau = -Fr$ karena perlawanan simpangan

$$\Rightarrow \tau = -F_{net}(r)$$

$$= -(P_{ud} - P_{he}) \cdot V \cdot g \cdot L \sin \theta$$

Hukum II Newton ^{rotasi}, yaitu $\tau = I\alpha$

$$I = mL^2 = P_{he}VL^2 \text{ dan } \alpha = \frac{d^2}{dt^2}\theta \text{ (sesuai dengan definisi)}$$

$$\Rightarrow -(P_{ud} - P_{he})VgL \sin \theta = (P_{he}VL^2) \frac{d^2}{dt^2}\theta$$

Karena sudut kecil, $\sin \theta \approx \theta$ sehingga $\frac{\sin \theta}{\theta} \approx 1$, maka

$$P_{he}L \frac{d^2\theta}{dt^2} = -(P_{ud} - P_{he})g\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\left(\frac{(P_{ud} - P_{he})g}{P_{he}L}\right) \cdot \theta$$

$$\Rightarrow \boxed{\alpha \propto -\theta}$$

(b) Dari persamaan pada (a),

$$\frac{d^2\theta}{dt^2} = -\left(\frac{(\rho_{ud} - \rho_{he})g}{\rho_{he}L}\right)\theta$$

dan mencocokkan dengan
 $a + \omega^2 x = 0$,

didapat

$$\omega^2 = \frac{(\rho_{ud} - \rho_{he})g}{\rho_{he}L}$$

$$T = \frac{2\pi}{\omega}$$

$$= 2\pi \sqrt{\frac{\rho_{he}L}{(\rho_{ud} - \rho_{he})g}}$$

$$= 2\pi \sqrt{\frac{0,17g \cdot 3,00}{(1,20 - 0,17g) \cdot 10}}$$

$$= 2\pi \sqrt{\frac{0,537}{10,21}}$$

$$\approx 2\pi \sqrt{0,0525954}$$

$$\approx 2\pi \cdot 0,2293370$$

$$\approx \boxed{1,440967 \text{ detik}}$$

7. $y(x, t) = 0,350 \sin(10\pi t - 3\pi x + \frac{\pi}{9})$

didapat

$$A = 0,350 \text{ m}$$

$$\omega = 10\pi \text{ rad} \cdot \text{s}^{-1}, \quad k = 3\pi \text{ rad} \cdot \text{m}^{-1}$$

$$\mu = 75,0 \text{ g} \cdot \text{m}^{-1} = 0,075 \text{ kg} \cdot \text{m}^{-1}$$

$$v = \frac{\omega}{k} = \frac{10\pi}{3\pi} = \frac{10}{3} \text{ m} \cdot \text{s}^{-1}$$

(a) Laju rata-rata ($\bar{P}$) adalah

$$\bar{P} = \frac{1}{2} \mu v \omega^2 A^2$$

$$= \frac{1}{2} \cdot 0,075 \cdot \frac{10}{3} \cdot (10\pi)^2 \cdot (0,350)^2$$

$$= 0,0375 \cdot \frac{10}{3} \cdot 100\pi^2 \cdot 0,1225$$

$$\approx \boxed{15,11283 \text{ watt}}$$

(b) Untuk mencari energi yang terkandung dalam satu siklus, digunakan

$$E = \bar{P} \cdot T$$

$$\text{dengan } T = \frac{2\pi}{\omega} = \frac{2\pi}{10\pi} = 0,2 \text{ s}$$

Maka,

$$E = \bar{P} \cdot \frac{1}{5}$$

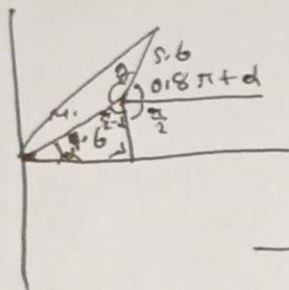
$$\approx \boxed{3,02256634 \text{ Joule}}$$

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12. $y_1(x, t) = (4,60 \text{ mm}) \sin(2\pi x - 400\pi t)$

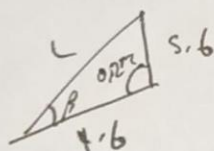
$$y_2(x, t) = (5,60 \text{ mm}) \sin(2\pi x - 400\pi t + 0,80\pi \text{ rad})$$

(a) Amplitudo gelombang resultan dapat dicari menggunakan metode fasor, yaitu dengan diagram berikut ($\phi = 2\pi x - 400\pi t$)



$$\theta = 2\pi - \frac{\pi}{2} - \left(\frac{\pi}{2} - \phi\right) - (0,8\pi + \phi)$$

$$= 0,2\pi$$



Menggunakan aturan kosinus, d. dapat

$$L^2 = (4,6)^2 + (5,6)^2 - 2(4,6)(5,6) \cos(0,2\pi)$$

$$\Rightarrow L \approx \sqrt{52,52 - 51,52 \cdot 0,80901699}$$

$$\approx \sqrt{10,8399494}$$

$$\approx \boxed{3,292331157 \text{ mm}}$$

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(b) Sudut relatif (berdasarkan diagram (a)) dapat menggunakan hukum/aturan sinus

$$\frac{5,6}{\sin \beta} = \frac{L}{\sin(0,2\pi)}$$

$$\Rightarrow \sin \beta = \frac{5,6 \sin(0,2\pi)}{L}$$

$$\approx \frac{5,6 \cdot 0,587785}{3,292331157}$$

$$\approx 0,9997762$$

$$\Rightarrow \beta \approx \sin^{-1}(0,9997762)$$

$$\approx \boxed{1,59966331 \text{ rad}}$$

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atau sekitar $\boxed{88,789167^\circ}$

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(c) Agar resultan maksimum, gelombang ketiga memiliki fase yang searah dengan resultan gelombang pertama dan kedua. Dengan demikian, berdasarkan subsoal (a) dan (b), didapat sudut fasenya $\approx 1,59966331 \text{ rad}$

$$\begin{aligned} \text{amplitudo} &= 5 + 3,292331157 \\ &= \boxed{8,292331157 \text{ mm}} \end{aligned}$$

14. $y(x, t) = 0,02 \sin(2x - 10t)$ dengan x m, t s

$$\mu = 1,9 \cdot 10^{-4} \text{ kg.m}^{-1}$$

$$\Rightarrow k = 2 \text{ m}^{-1}, \omega = 10 \text{ rad/s}$$

$$A = 0,02 \text{ m}$$

$$(a) \lambda = \frac{2\pi}{k}$$

$$\begin{aligned} &\Rightarrow \frac{2\pi}{2} \\ &= \boxed{\pi \text{ meter}} \end{aligned}$$

$$\omega = \frac{2\pi}{T}$$

$$\begin{aligned} \Rightarrow T &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{10} \\ &= \boxed{\frac{1}{5} \pi \text{ sekon}} \end{aligned}$$

(b) Laju rambat gelombang dan tegangan tali:

$$\begin{aligned} v &= \frac{\omega}{k} \\ &= \frac{10}{2} \\ &= \boxed{5 \text{ m.s}^{-1}} \end{aligned}$$

$$v = \sqrt{\frac{F}{\mu}}$$

$$\begin{aligned} \Rightarrow F &= \mu v^2 \\ &= 1,9 \times 10^{-4} \cdot 25 \\ &= \boxed{4,75 \times 10^{-3} \text{ N}} \end{aligned}$$

$$\begin{aligned}
 (c) \quad v &= \frac{dy}{dt} \\
 &= \frac{d}{dt} 0,02 \sin(2x - 10t) \\
 &= -0,2 \cos(2x - 10t)
 \end{aligned}$$

Maka

$$v_y(x,t) = -0,2 \cos(2x - 10t) \text{ m.s}^{-1}$$

$$a = \frac{dv}{dt}$$

$$\begin{aligned}
 &= \frac{d}{dt} (-0,2 \cos(2x - 10t)) \\
 &= -2 \sin(2x - 10t)
 \end{aligned}$$

Maka

$$a_y(x,t) = -2 \sin(2x - 10t) \text{ m.s}^{-2}$$

15. $\text{---} L_0 \text{ m}$

$$0^\circ\text{C} \Rightarrow 331 \text{ m.s}^{-1}$$

$$\text{Nada dasar } 261,7 \text{ Hz}$$

$$20^\circ\text{C} \Rightarrow 344 \text{ m.s}^{-1}$$

(a) Frekuensi nada dasar saat $T = 20^\circ\text{C}$

$$f = \frac{v}{\lambda} \Rightarrow \frac{f_2}{f_1} = \frac{v_2}{v_1}$$

$$\Rightarrow f_2 = \frac{v_2}{v_1} \cdot f_1$$

$$= \frac{344}{331} \cdot 261,7$$

$$\approx \underline{271,98 \text{ Hz}} \text{ (sekitar } 272,0 \text{ Hz)}$$

(b) Mengukur perubahan, didapat

$$\frac{v_1}{L_0} = \frac{v_2}{L_1}$$

$$\Rightarrow L_1 = \frac{v_2}{v_1} \cdot L_0$$

$$= \frac{344}{331} \cdot L_0$$

$$= \left(\frac{344}{331} - 1 \right) \times 100\%$$

$$\approx (1,03927 - 1) \times 100\%$$

$$\approx \underline{3,927\%}$$

$$\Rightarrow \% \text{ pertambahan} = \frac{L_1 - L_0}{L_0} \text{ (atau } \approx 3,93\%)$$